

Lance Da Costa ~ Active Inference Insights 010 ~ Mathematics, Time, Selfhood

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hello everyone and welcome back to active inference insights as always I'm your host darus parvey Wayne and today I have the great pleasure of speaking to Mr although very soon it will be Dr Lance D Costa Lance is a PhD student with Greg pavus and KL friston jointly at Imperial College London in UCL and his work lies at the intersection of mathematics cognitive science and machine learning indeed he is at the very Cutting Edge of the technical formulations underlying AC of inference and so I'm personally looking forward to learning from him today and I hope you are too ladies and gentlemen I bring you Lance the Costa Lance thank you so much I I genuinely meant what I don't really like doing the introduction because it feels so you know sort of a bit stagnant and rigid but I genuinely meant what I wrote there which is that I'm really looking forward to learning than for having me and appreciate the introduction yeah um and you will when when is when is one when do you expect to become a doctor when's that total happening so my defense is at the end of February so oh my gosh it's inth yeah wow fingers crossed I'm sure you'll smash it thanks so before we start getting into the maths I this is very exciting for me because you're the first I would say pretty much pure mathematician we've had on we've had computer scientists we've had cognitive scientists but no one really uh delving into the the nether regions of the maths um you know when I was researching what you're interested in I came across um the fact that you're interested in World models and I'm also very interested in World models so I wanted to start there as's like a teaser before we get into the maths and I wanted to ask whether you could explain what you mean by a world model and how you distinguish that from a self model and the reason why I ask that question is kind of twofold one is that I think the word world models might be slightly misleading because it kind of implies that we're modeling the world and in some sense we are but we're definitely not modeling the entire world so it might be called something like an econ Niche model yeah although that's less catchy and then the other reason is I think there's still an open question at least to my eyes in active inference about whether we

are modeling the world and ourselves or whether we're just modeling the world and we fit into that in some way right so it would be great to unpick that if possible yeah that that's actually a Cutting Edge question but I really like it great so let me start by world what I mean what I personally mean by World model is that you as an individual you have always some sensory data coming in off active Visual auditive and so forth and you're going to build a model for what's going on outside that causes your data um now the model that you build is a generative model which means that you can generate kind of factual data about what might happen in the future what might happen if I do this and that uh it's a probalistic model because you're always uncertain about what's going on around in the world and it's also a causal model um this means that you're not um you're not exploiting the well you're not exploiting just the correlations in the data that come in but you're actually explaining the causes of the data something like more powerful than than a statistical model uh now I think you're right in calling it an eon Niche model because I mean the data that we have coming in at any point in time is just from around us and so that's ultimately what we want to model uh when we talk about the free energy principle we say that the brain optimizes its model to maximize accuracy and minimize complexity um in that and and that entails uh having a model that's as simple as possible and therefore you don't want a model things are like too far away from you important from you and so this is why having an eon model as a terminology makes more sense sure so when you talked about world model and self model and this is where it really gets Cutting Edge uh so in the like simple applications of active inference and AI you just care about modeling the world um but now as humans we don't only do that so for instance if I take a particular if I make a particular decision or take a particular course of action and you ask me why did you do that I'm not going to be able to uh answer in saying oh this neuron did this and this neuron did that and this happened but I'm going to give you a high level overview using natural language of why I did this now this uh is the idea of metacognition which means of having a model of the self in addition of having a model of the world so you have a model of yourself and you have a model of your own cognition and so this is kind of like a higher form of cognition and active inference uh which has not so far been really implemented out there I believe at least in machine learning it's kind of come out in papers on um modeling meditation with active inference L's work yeah I like yeah yeah yeah very nice cool that's really so so you so this is yeah this is a question I've been entertaining when I wrote this first paper on Flow that I've now mentioned far too many times so I shall mention it again um I was thinking exactly what are we talking about when we're talking about something like an epistemic agent model and uh

I remember one comment commentator said oh is it just metacognition so that's an interesting alignment there um because in a sense yes you're right we have to reflect on our own capacities and model that as an inference my only question there is you know in L's paper you have these really beautiful hierarchical Baye graphs do you think there's a there has to be an upper bound on the computational complexity of that metacognitive and what and is is that just a limitation of the computation in and of itself or does it mean that there might be something ontologically sort of actually there right what what do you mean by ontological exactly uh let's well let's use a word well I'm not going to invoke something like a Cartesian soul but is there any I guess is there anything unique beyond the algorithm beyond the probabilistic model right uh so what I would say to that I mean what we're personally investigating with Lars is what does metacognition actually allow us to do allow us to do that's uh you know important for survival that an agent that does not have metacognition is not able to do so the idea is if metacognition arose at some point is because it increases our chances of survival our fitness it minimizes our free energy um how exactly I mean how is metacognition useful beyond what you know what we just talked about uh is currently unknown at least for me sure yeah I think it's I think it's a complex issue because it it's entangled with common folk psychology Notions of what it is to be a self um so most people will think of themselves uh probably if you ask them they would say yes I am this substantial thing somewhat behind my eyes uh that persist through time so somewhat of some kind of cartesianified ego and I actually found I find that active influence is a lot easier to pass once we take that out of the equation when you just have the physics or when you just have the statistical mechanics it becomes a lot easier to say okay maybe this is maybe the self is just a further inference that's kind of where I was pushing you in terms of the ontological thing because I think people listening if they haven't really you know thought about the the Buddhism and the metacognition and the Derek parit related philosophy might think well there's just something there that's doing it right there's something there that's a homunculus behind the eyes that's actually doing the calculations that's doing the variational basing inference but I assume you can't you know there's no notion of that in your models um so you mean like kind of like an agent inside the agent indeed and obviously this leads to a homuncular infinite regress because who's watching the screen and so you know and so on but I think people have an issue with just a raw notion that there's just algorithmic computation going on because they're going to ask well who wait what who is doing it and that leads us back to some kind of uh active agent that's actually there not just an emergent phenomenon right uh so so I have a few things to say

about that I think uh so when we when we engage in meditation we are computationally looks like we're building a higher level in our giant model which is kind of like another agent inside that makes inferences about our cognition yes and so when you're a trained meditator you not only think about the world or experience the world but you experience your own cognitive processes and you train yourself in doing that now how that looks like computationally is that you have a hierarchical gen model with many levels of of abstraction but when you translate that into math and physics it's kind of this idea of the Russian doll yes like an outside agent and some more like more agents inside hierarch stacked that just infer each other yes Mark of blanket in some sense yeah nested Mark of blankets and so but that but you know that analogy implies that there is some outer shell that might be right only that might be you know yeah this makes me think of Maxwell Ramsey the M's paper on Consciousness that there's this right only layer that is contextualizing but not contextualized um but that's kind of the other way around right that's like that's the that's kind of like the prime mover unmoved whereas you I feel like your argument is saying well everything is contextualizing everything else and actually what we might call the AL shell the autobiographical self is just the aggregation of all of these other selves does that sound about right yeah it I feel it's uh kind of hard for me to to describe it um and I'm also thinking along these lines but I would say I mean the freer principle everything is dependent on the blanket that you select right and is the idea that we have multiple nested blankets now the story that you tell depend on which blanket you decide to talk about do you I think people have an intuitive folk psychology notion that of course there's a point at which the modeling of me as a discrete identity needs to stop St so we could keep Mo so so most people probably think about the surface of the skin the epithelium and say well that that that's me right and then we can get into conversations about extended mind hypothesis but I actually don't think it's that relevant here I guess the point is um it seems like that the point at which we stop the mark of blanket is actually it's not arbitrary but it's an open question it's philosophically vague so to speak um which which which negates the kind of folk psychology notion that there is something like me which has these strict bounds yeah so so I I completely agree with kind of like the dissolution of those straight bounds once you start looking at it very carefully intuitively I think that my you know my hands are mine because I can control them in a relatively precise way and I think this is where that comes from okay it is the preciseness at which you can act on something or inert that's nice yeah when you when you do a lot of meditation or when you when you look at it like very carefully you start realizing that there's so many things in your body that you cannot control it's kind of obvious

when when you think about it and you start realizing that you have very little agency over many things that are happening in your body so I think everyone of us has things that they're satisfied and they satisfied about themselves when you think about it where maybe I'm dissatisfied with something about myself uh but then this is not really my fault of me being that way so here's an example of um of something that I cannot control once you reflect that on that more and more you start realizing that you know uh we are self-organizing systems just like a tornado is a self-organizing system and we have a lot of things that are going on inside our bodies inside our minds that you cannot just cannot control H so yeah that's interesting I guess I guess my rebuttal to that would be arguably the literature says that or suggests there are processes that underly something like a uh minimal self like inter receptive inference so that can be done reflexively or through kind of alistic control so let's say my glucose level drops Beyond a certain level I either you know secrete the necessary hormones or go out there and eat but let's just say I'm secreting those hormones I'm not act you know my autobiographical self doesn't have the doesn't have the sense it's doing anything yeah th that mechanism of prediction Eros suppression is supposedly at the heart of the sense of mindness and presence so how do we resolve that if we're casting selfhood as that which is within my control um well I I don't know to be honest but it's a it's a a difficult question but uh yeah I mean the the more the more I think about all of this and the more I read the more I realize that we have control over very little yeah and so in in the meditation books so I keep coming to that because they're a source of inspiration for me for describing high level cognition there is this um like fundamental truth qu quote unquote which that there is no self the more you start like investigating the Mind through self inquiry the more you realize that there is no self yeah yeah um so maybe it speaks kind of to the things that we're just discussing sure yes this is yeah this is an interesting point um I guess you one would have to ask very dedicated meditators uh whether when they're doing let's say Vasa and they break that kind of Duality um because you know the subject OB object duality yielding to Oneness is kind of a seminal phenomenon that occurs whether that sense of mindness and presence is also lost so there's an you know uh John vvi who's one of my favorite cognitive scientists and who was on the podcast speaks about adverbial qualia and what he means by that is you there are these so-called pure Consciousness events and even in those the sense of presence and perhaps mindness although we kind of weren't sure um might still be present which I which to my ears sounds like some computational mechanism like interceptive inference still being at play so I I guess this is calling for calling for a denial of the strict distinction between the body and the mind that the

interceptive inference can still underly or bodily inference can still underly very important phenomenology yeah well uh active inference is definitely taking this uh philosophical position that cognition is embodied so so yeah it speaks exactly to what you said now I I just want to add something else so we talked about the the model of the self and I would say there's actually two models of the self so there's the model of the body and and this has a like clear um evolutionary like reason for being there so for example if something in my body hurts I need to figure out what to do about it for it not to hurt so it's about preserving your homeostasis and fulfilling preferences now there's this other level that we've also been discussing which is like the model of our own cognition these are two different things yes that's that's interesting yeah cool there's lots there's lots to chew on there I'm glad that I'm glad that this is Cutting Edge questions that need Cutting Edge these are really things uh that I know a lot of people are thinking about spend yeah I think I think I'm gonna I'm gonna have lws on at some point yeah he owes me that to be honest uh he keeps promising okay um let's to the bet you know this is this self-modeling philosophical conversation is more my bag but let's take that deep PL uh plunge into the mathematics so as a as a kind of a forewarning or a precursory warning to the audience I'm not a mathematician um so everything I'm doing when I'm reading the math I'm trying to ground in intuitive philosophically robust Concepts so just just for the audience and yourself please do excuse me if I [] **up um you won't I mean this is a I think it's very important to be able to talk about these things without the math it's literally impossible to talk in math so yes yeah yeah yeah there's this about this in a in a natural language way that's hopefully as simple as possible please well yeah I guess that's how it has to be taught is through metaphor um sort of yeah life through metaphor for George Lov kind of stuff so um start starting right at the basis in Act of inference the world is described as a random dynamical system defined or described by a stochastic differential equation like a leran equation yeah why why is it worthwhile starting there why is that the starting point right um should should I explain briefly what Lan equation is perhaps sure it'll be better you doing it than me okay uh sure well um so here's the thing we want a fundamental theory of the brain and cognition and we want it to be grounded in math and physics um now if you look at the fundamental theories that exist in physics these would be uh general relativity classical mechanics statistical mechanics quantum physics they are all based on stochastic differential equations or long my equations at some level now um a stochastic differential equation is a very fundamental and also basic object and I'll explain why it's basic it's basic in the sense that when**

you're modeling a system there are the known knowns and they're the known unknowns yeah stochastic differential equation uh brings the two together so there may be the known knowns about the system which could be forces that act on different components of the system this is going to drive the system deterministically along a predetermined path and then there are known unknowns these could be random fluctuations for example if you have a drone this could be random fluctuations due to wind if you have a boat random fluctuations due to waves and these are things that you possibly cannot accurately capture so you're going to summarize them in a random term and then together so you have this deterministic uh term that gives you a predetermined traj and you have this random term that sort of perturbs the trajectory in a random way and when you put the two together you get kind of like a random evolution of your system with some structure and this is basically what the Lan equation or stochastic differential equation um encodes so this is to say uh it's a very general starting point and it's also the starting point that's consistent with the rest of physics and that's why we start there when we uh when we think about the free principle excellent yeah so there's a couple things I wanted to pick up um am I right to say that it's about the rate of change of so of some X let's say of some states is that is that so that's that's kind of what we're looking at here is the okay and then the second thing is um this notion of random noise if you speak to certain philosophers for example they might say well it depends on who you ask I guess um because for a universal you know for um for some Demon that knows the entire unfoldings of the world in a single glance there's no Randomness true and yet to us the modelers well it's just computationally impossible to know every single element of the wind let's say or the waves so what what are we talking about are we talking about Rand Randomness in in and of itself in the actual system that's being modeled or for the modular this is a great question uh so here it's it's agnostic in the sense that uh you could imagine you could model the the the system that you're trying to model in all in explicit detail so that you know absolutely everything that's going on in that case you would have no Randomness in that case you still have a stochastic differential equation with no stochastic term which is an ordinary differential equation but still atic differential equation so the same math apply sure you just have a random term that's like zero yeah yeah yeah maybe there are things that you cannot model accurately in the system so you put them in some noise and the same math applies so I would say it's noise from the perspective of the modeler or not and and we don't actually care the point is that once

you start from the assumption that you can model the system in that way then then things carry through and I make the presumption I'm presuming here that that noise is Gaussian is considered Gaussian it's consider a normal distribution gem yeah so generally for uh for for you to be able to do mathematics you take Gaussian noise is that an issue well it depends on who you ask like uh most math and physics is based on Gaussian noise because it represents the most the world most accurately you have something called the central limit theorem that if you have a lot of like fluctuations they kind of average away in a pile of sand yeah yeah so your noise turns out to be often times very well approximated by by Gaussian process uh but then there there are instances where Gaussian noise is not accurate so for example if you're talking to people who model the financial Market they do not use Gaussian noise they use noise with like jumps to model like a very sharp um yeah jumps in the market so Gaussian noise is a good place to start I think ultimately as we work on the free energy principle mathematically we want to be much more inclusive sure isn't that I mean just as a side isn't that a remarkable fact that like stochastic processes tend towards Brownian it's just like yeah I think it's awesome it's amazing yeah yeah I mean you have all these remarkable properties of the Gaussian distribution um and it's also like the only one distribution that we actually know how to work with very well so it's a good thing yeah it's cool um and then I guess my other question about the Langevin equation is you have sort of $f(x)$ as the deterministic force so it's a function of the states of X the flow of X is that right yeah right yeah but it doesn't have a kind of comma T next to it um why is that not uh a function of time as well so it could be in its most general form F the flow would be also a function of time now if you think of it let's say you have a state space and you have time you can just incorporate time as another element as another state so you consider your state space to be space time and then you're back at $F(x)$ right so I think that's why that's why often times people just go with $f(x)$ cool okay excellent now let's jump to the next step we have density Dynamics and the Fokker-Planck equation and the path integral formulation right very fundamental question what is the difference between those two there there's no difference there's just different ways of articulating the same thing okay so the Langevin equation gives you like the rate of change as a function of the flow and a random component random noise yeah uh then when you sort of solve that you sort of get um this is the probability that the system T takes this trajectory and this is the probability that gets this other trajectory and so forth this information is what's carried in the path integral formulation the path integral

formulation gives you the negative log probability of a path y and then you the focal plank equation uh which says at any point in time this is the distribution that encloses the system so you might in a predefined place that you know I mean the system might be might start in a particular location in phase space m because of the flow and the random fluctuations uh at time $t + \Delta t$ the system will be it could be here it could be there it could be there with a different probability and then there's a prob distribution that summarizes this and this is what's including the focal plank equation FOC plank they both come to the same conclusion you run the maths both ways and they all converge the answer that it's it's exactly the same thing yeah it's like different ways of looking at the same thing and depending on the question that you want to answer uh taking one formalism or another is easier but the point is that you can always go from one to the other and back nice okay that's cool um okay let's jump into the Fokker-Planck equation because I've been I've been doing some reading and I might be wrong but I'm excited about it it's excited me so cool yeah so from what I've learned it's a partial differential equation yeah that rate that relates the rate of change of the probability density function so that's that P with the little dot and X comma t with respect to time drift and diffusion right how's that sounding yeah it's a great summary I can elaborate on that a little bit to please well people have no idea what I'm talking about so okay well you'll see that you got it perfectly right oh thank God for that all right please explain in a 1 equation you have the rate of change of the state y the flow and the random term yes um now this is what physicist called it but if you talk to mathematician he would say you have the rate of change of state States the drift and a random term which they call the diffusion yes nice or which is characterized by the diffusion so then the focal plank equation is very similar you have instead of having the rate of change of the state you have the rate of change of the probability density that describes the state and then this is given as a function of the drift so the flow and the diffusion so the random term Okay cool so so um if we were going to cast this in like in terms of just cartesian coordinates just X and Y axis would it be right to say that we could have X as like the position let's say of because I I read there's a lot here in sort of brownian particles so let say I had my particles in some you know doing brownian motion the X would be the position of some particle and the Y would be the probability of finding it there yeah when you talk about the focal plank equation yes yeah so the focal plank equation would be that so at each point in time the the solution to the focal plank equation is the density of the system that evolves

over time and so you could plot this on the XY axis as you said on the x axis you would have the position of the system on the y axis you would have the probability distribution and so if we imagine things are simple so if we imagine the system is always gaussian distrib according to gaussian distribution then if you solve the FOC plank equation you would have a gaussian that Evol over time Su the mean would be shifting and the variance would be shifting also potentially nice okay cool that's that's really useful so how would we then integrate time is it is it this like is is the is that X and Y AIS considered like a snapshot in time yeah okay excellent yes okay cool that's actually really useful because I think people might got confused by that um yeah now let's get to so this gives us density Dynamics but when we talk about active we about we talk about steady state densities right so people I think have this intuitive notion from whether they're coming from dynamical systems theory or something else of a kind of basin right in ATT tractor set I mean I think of it kind of like a magnet um metaphorically what is how do we get from density Dynamics to a steady state density um so I should cave at this by saying that the latest formulations of the free energy principle are not based on having a steady state density that's like a new thing and I think for like intuition it's best to think about the steady state case okay so um kind of like the intuition is that if you're modeling an agent uh it has a preferred set of states that it likes to come back to or that it like does everything it can to like be back at those States you could think of like homeostatic States for instance um you can also think about the fact that if you're modeling A system that does not dissipate in time so let's say like not a gas for instance uh then the system should also have like a steady state at least over a short period of time because it doesn't dissipate right so now we're going to try to model um I mean the free principle is about modeling or analyzing those systems that have a steady state that have these sort of preferred set of states and so you ask if a system has a preferred set of States and it's also decomposed into an organism its boundary and what's not the organism again this could be like a living organism it could be a stone it could be anything that has a boundary uh then you basically look at what the internal states do what do this what does the boundary States uh do how do they evolve and what do the external states do and then the conclusion is that in general the internal States look as if I mean I mean they they're you can describe them as inferring the external States based blanket States yeah yeah yeah yes that's an interesting formulation and one we could come back to which is this kind of uh yeah tracking in some variational Basi and inference way um we've spoken

about this sort of looking like very important two words in the free energy principle um a number of times in this in this podcast but I was curious more about in terms of the fuer plank equation I don't know why I've turned into a um so so this is the way that I would think about and please feel free to from what I've learned very you know in a very shallow way is that we had this um this probability density function so let's say as you say we have our X we have our position y we have our probability distribution let's just say it's gaussian and per attractor State per organism per attractor State it's going to be different right so mine might be really close to this size yours whatever um what what I kind of understood as a steady state is over time so let's say now we like put all of those images of the X and Y axis on top of each other right right and let's say they were somewhat translucent you wouldn't if it's steady state you expect that probability distribution to like stay on top of itself if it's not you'll expect it to shift from side to side yeah or or like become completely flat all okay all become completely flat I guess reincarnation would be kind of gaussian flat gaussian what what do you mean by reincarnation sorry well let's say I it was it was a tongue and- cheek comment but let like I have a gaussian distribution of being in a certain state right right right I die I enter into the EPA I dissolve into the heat I flatten out and then I reappear as a frog and I enter into a new gaussian distribution yeah that's a good way to put it all right nice okay so is that is that does how does that sound mathematically what does that look like mathematically okay so um I think one yeah I think there's kind of like two scenarios uh one one is that there's no steady state and the system dissipates um and therefore you're you're like snapshots of uh of the like the density at each point in time will converge to a flat distribution okay um and and the other case is that there is a steady state and uh the snapshots converge onto like a a clear well defined distribution yeah which be like a gaussian or something else and the is that the point the point of a steady state is uh that if you start the system in that distribution it will remain in that distribution sure although okay okay fine and and that's strict as in you will stay within that distribution or is there permission let's say in sentient creatures like our own for that distribution to change somewhat um so that's a good question uh there's kind of like two answers to that you being at a pre-specified distribution does not mean that you're not doing interesting things yeah so if we just think about a convection cell you basically like or or a tornado you have like this circling around uh the probability distribution of where the gas is or where the turning is just stays constant it's kind of like this ring shape uh but you still

have some interesting motion inside um right so I might not always doesn't me right because I might just you know over time it will look like I'm converging on that state but I might not always be there yeah your probability it's kind of like hard to wrap your heads um if you think about like homeostasis right if you model if you like let's say temperature if you model temperature you can imagine it would have a gaussian distribution sort of with the Peak at 37.9 degrees Centigrade let's say but you are not going to spend your entire life at 37.9 degrees Centigrade exactly you're going to move around and on will be according to this Gaussian distribution right right okay cool um okay good um I I want to jump to that notion of the internal States tracking the external States cuz I think people struggle with that um and I definitely did for a while so so the way that I read internal and external is tracking and tracked um that for me has been a really parsimonious and useful way I think I got that from Maxwell that that's a really nice way of thinking about it because the internal external is such a philosophically Troublesome distinction so tracking and track and then I think of something like the equation variational free energy and I think about the KL Divergence between the recognition density and the true posterior and the maximization of model evidence now in that I guess I'm not I don't have to necessarily make this strict distinction between the internal States my States and the worldly States because all I'm doing there is I'm saying I have you know give uh given some observation I have a probability distribution of some State whether they're mine or whether they're the world's because for me I'm not going for me there that distinction is not that strong because for me I think you could pose that actually all that we are are just this aggregation of worldly States right I am just I am one temperature distribution I am one glucose distribution that's not mine that's the world's yeah um so so that's the first thing right so we're doing this x uh X give and Y this recognition density and then you have the maximization of the model evidence which is the probability of why given your model but again there's nothing like internal that's going on there it's just statistics so do you that do do do you think that it's somewhat Troublesome to cast like put this little homunculus or this little ego in the internal States and being like he's doing sufficient statistics is that do you see that slightly Troublesome um I think it's a nice picture for ination but it's certainly uh it can be misleading for the reasons that you pointed out uh internal and external States can be interchanged yeah and again it's all about the blanket yeah the external States could be part of your own body and the internal States could be part of your brain and

the blanket somewhere in between or the internal States could be like a school of fish and the external States could be like a predator right um yeah I just never thought I this is no attack on anyone I don't know who came up with it probably KL sorry KL I never thought that was a particularly good formulation of the free energy principle I always thought that um you know things that seem to persist look like they're maximizing model evidence for themselves is way better um but this is something to take out with him yeah but but it's kind of like two things we're touching on here because tracking and the Mel evidence uh they kind of like two things are going on because when you think about it the free energy is this K Divergence plus surprise uh and we use this tracking to like minimize the free energy right so what I mean by that is the maximization of model evidence or I think it's the minimization of log model evidence but it's the same thing um that entails that in of itself what the way I'm trying to frame it is that that entails some form of tracking yeah because you've got that you got that probability of why but it's sort of given your model um so yeah I I just think for people this is again this more an education point about active inference not about the maths or anything um I just sense that for me it's always been more intuitive to think about that although again it's hardly intuitive is it I mean it takes a bit of time to really wrap your head around yeah there's there's another important point that came out in the research uh when we are talking about the fact that you can I mean a lot of people say that you can always interchange uh internal and external States and that one is tracking and the other is tracked but it turns out that this is not always the case because for the internal states to track the external States they need to have enough like representational capacity make inferences about all the internal states are out there they don't have enough representational capacity maybe they might just infer some or infer linear combinations of external States in any case they're just going to have like part of the picture okay that's no that's that's interesting which means that for external states to track internal States they also need to have like enough representational capacity and a sufficiently high quality coupling with their own sensory states with the boundary to make inferences about the internal States so you always have this synchronization and this inference going on you Cann necessarily say that the internal States will be tracking all the internal States all the time or or vice versa right kind of depends on the Integrity of the coupling and how much representational capacity these states have well again that's why for me this maximization of model evidence just seems a little bit simpler because whether you're a drop of oil or a rock or

human we can sort of arbit you know not arbitrarily but we can generously give out models for each one of them and say that they look like they're maximizing model evidence yeah now um like how long something lasts is a different question um so my you know the the the salt that I put in water maximizes model evidence for itself being so for a very short amount of time so you know you could cast that as it surprise will being very high but at least for while it's salt it's maximizing model evidence and then I guess it's maximizing model evidence within a Markov blanket that's the water salt solution anyway it's an interesting foray but it's no that's a good point yeah we're now going to jump to the next part of this so people who want to follow along by the way a lot of this is from the free energy principle made simple but not too simple or I think that's the name of the title that's right yeah that's a great this this is a well I wouldn't say it's necessarily a perfect starting point for someone who hasn't done maths or physics don't we I would I would recommend um the textbook so pariet Zulo and Friston you got recommendations actually yeah there's also Maxwell's paper on Basin mechanics yeah that's very good and off beliefs and this is really cutting edge in describing the math but it's also meant to be much more approachable yes okay that's yep so Maxwell's 2023 paper which is on the uh pre-print server um the textbook and I've also been reading and we'll come to this I've been reading sanjiv Nam joshi's draft of his p uh book it's it's it's being sort of distributed there's a group in the active inference Institute where you can read it and that's been really useful so there's stuff out there and and there podcast and so yeah but if you wanted to follow along with this this is that uh B uh free energy principle made simpler but not too simple so the next thing that you you come to there is solenoidal and gradient flows right so maybe we can talk about maybe we should start by just explaining what those two flows mean and how they're distinct from one another okay um so this is a the composition of the flow in the Lan equation yeah and so uh to put it simp to put it very simply the um well just to give an analogy in Vector calculus you have this fundamental theorem about that's known as the hamle dec composition yeah which says that if you have a vector field so think about a bunch of vectors sitting on a table uh you can decompose them onto uh one that's kind of like circling around an origin to put it simply and one that's going inside or outside okay so mathematically this is what we're doing here to decompose uh the flow in the L equation now what this really means in practice is that you have uh the solenoidal flow which is the interesting part of the flow and the how did you call the other one uh what did I call it gradient flow right and there's a gradient

flow that's the in uninteresting one if you are um if you are a living system you have by definition solenoidal flow and why is that solenoidal flow is is um related to time irreversibility and entropy production um okay if if you don't have solenoidal flow it means that the system if you if you play it in time uh as time goes forward and then you you reverse the movie statistically the system will look the same so it's essentially a dead system it could be like a stone for instance yeah on the other hand if I you know you play the podcast for 20 seconds as time goes forward and then you play it backward for 20 seconds it will look super weird and very different right this is the proof that there is time irreversibility time has a there is a direction time um and this is yeah characteristic of the fact that we are alive interesting okay so let me just get some some more you know some simple things outlined here so you said the hel holtan the Helm Holt's decomposition is of the flow is that if I'm looking at the long Jaan equation that's not the that doesn't include the random noise that's just $f \circ f_x$ f yeah it's a de composition of $f \circ f_x$ but also done in relation to the noise term so is the noise term so would that what I'm asking here is when I'm splitting it into solenoidal and gradient is that splitting just the $F \circ f_x$ or the $F \circ f_x$ and the yeah Okay cool so okay so there's a there's a dissipative aspect in all of this so the analogy that I thought of was um if you create a Whirlpool let's say in your sink or in a cup of coffee you get this solenoidal flow right it's conservative but due to I presume just friction you get this dissipative uh ele this this aspect where the swirling motion Fades away yeah totally right let's is that is that the gradient flow then sort of infiltrating a purely solenoidal flow yeah that's absolutely right another example is if you put milk in your coffee yeah there will be like the standard mixing uh of things which will be the gradient flow then if should start stirring the cup of coffee with a spoon then you're adding C flow and is it where they meet that Things become like we can say things are dispa of because for me philosophically dispa of means that you had order and now you've got less order but if you if everything is if there's no solenoidal flows then arguably there's no is there is there any order to begin with it's hard to say like I have yeah I have trouble relating these Notions to order but I think there's like there's like an indirect way of relating them which is so Noto flow is always something that circles around something yeah you think about the Circadian rhythm yeah us waking up and going to sleep all these are Cycles they are examples of sidal flow an aof flow has um produces entropy so I think that's in some sense related to order and disorder if you want to maintain as a living being if you want to maintain cidal flow or just as a physical

system you need to be using energy and dissipating entropy so that's kind of the idea and that's just a consequence of the mathematics and the physics yeah yeah basically okay the fact that we have senoidal flow within us like rhythms and Cycles means that we need to eat and uh and secrete right right okay okay because in some sense the that's a form of order which is neg entropic yeah yeah okay now uh okay so what is this um the other thing that comes up is the term Divergence right so people say that solenoidal flows are zero Divergent yeah what does that what does that mean right uh it's kind of that's just kind of a mathematical term uh but uh because if you take the Divergence of the vector field it's zero but that's getting to that I'll give you a picture if we if we are at our steady state distribution so let's say um our system is two dimensional and it's distributed according to gaussian that's sitting on my table here um the soal flow is uh motion around the Contours of this gaussian distribution and the gradient flow is motion towards the peak or away from the peak interesting and so we have our overall motion which is given by the flow in the Lan equation which could be in any direction and then um the fact is you can decompose it on motion towards the peak so towards states are more expected or preferred and motion away from the peak strictly away from the peak these two are Divergence free flow you can also the other part of the decomposition is motion that just goes around the Contours of the distribution which means that it does not approach the peak or go away from the peak is it if I'm thinking about it diagrammatically does it matter where that circulation is going does it so so it could be at the base of the gaussian which means it would in some like again just diagrammatically means it would be wider or it could be at the top of the Gaussian which means it's occupying these high probability states does it matter um I think it matters as a signature of the system probably I mean this idea of Coto flow is one of the most important like signatures of uh of being alive and what your states are doing what kind of Cycles you have in your system or in yourself uh so it's a very important um descriptor of what's going on okay cool and the other thing was um and again you can tell me when I'm when I make a terrible fuxa is that solenoidal flows break what's called detailed balance yeah and what I've learned about this is that detail balance is when the rate of transitions of any two states is equal in both directions that's right so like you can think of it in terms of I I I thought of it in terms of like a crowd going in and out of a like a stadium or in and out of a room that rate is going to be the same in and out and that I believe and again you can tell me if I'm wrong there's something to do with boltzman boltzman's h

theorem which says that that that detailed balance implies zero in that entropy right um so so two parts um it's absolutely right that detail balance is means that there is flow um in and out say of a stadium at at the same rate it means that if you play the flow in and out of the stadium forward or backward it will look the same there's no production of entropy and there's no solal flow and the system is time reversible yeah yeah yeah okay so that kind of explains why the it's sort of N Net Zero entropy right if entropy is foundational to the passage of time yeah yeah yeah basically yeah as soon as you have S flow time uh starts taking place you start having a narrow time okay so if if we took so yeah okay so this is interesting so let let's let's kind of this is an odd exercise but we can kind of build a human from scratch right so let's try right um so the the gradient flows are going up or down the probability distribution so we take a human as just our gussian distribution it's kind of going up down but let's say it without any solenoidal flows we have this detailed balance yeah so it's going up and down now does that not mean that because it's doing it uh equally you're going to end up at that Peak uh no because you have equal flow towards the peak or away from the peak so how does that average out what does that look like in terms of where I where the crowd might be or they're just everywhere are we talking about the human or or the crowd going in and out of let's say the crowd let's say because we've got detailed balance so let's say the crowd okay okay if I'm gonna say okay uh I have a random friend X and they're they belong to this crowd that's uh behaving in terms of detailed balance and I want to find him is there a probability distribution that could help me to find your friend yes who's going in you know is part of that TI um yeah I'm not sure about that okay what I will just say is that like if you have one exit in your stadium with people going in and out at the same rate yeah and they're kind of like mixing with each other so who is who you have detail balance you just having one entrance and one exit then you do not have detail balance because you have people coming in one way and going out the other way so you start having a cycle ah okay now that's that's okay that's really useful nice so let's now talk about the human in a sense because the hum a combination of we're kind of what Carl would say we're in this goldilock zone so we're not orbital planets with just solenoidal flows so we so we're not just this cular uh circular thing but we're also not just NE entrop non-- entropy one it right so we have both so we have both yeah so yeah it's a very interesting sweet spot if you just consider systems that have just the gradient flow yeah are dead and you know they might look like gases and stones and stuff like that yeah

yeah yeah if you just look at things that has tornado flow well I would say they're probably dead as well it could be like like planets orbiting around the Sun like they're going One Direction and not the other and we are kind of like in between where we have both going on so I guess this is where my question was going which is okay let's say let's take a Rock The Rock has uh just gradient flows so it's dead but it seems to me that the rock is doing a pretty good job at self-evidencing right so does in some ways the solenoidal flow take us away from Yes it gives us animacy and whatever but does it take us away from like just being being able to rigidly self-evidence rigidly maximize model evidence no um no no it does not and so the beauty of the mathematics here is that we're taking the most General sort of equation that can describe the world so the F principle applies if there is C Noto flow and no decied flow yeah it applies there is decied flow and no solenoidal flow and it applies if there's both yes but what I mean by this is it's it's probably worth mentioning that I'm kind of thinking about this over time so my uh my plastic bottle is going to outlive me and and because things over time have to you know have to abide by the free energy principle they have to look as if they're maximizing model evidence or doing some variational Basi and inference you could say if we had some normative sense which again I'm quite skeptical but you could say like the water bottle is doing better self-evidence over a period of time uh yeah over over a long period of time certainly and that's because well that seems to me to map on quite nicely to solenoidal gradient flow distinction because like if rocks and uh I don't know particles are not solenoidal where where am I I feel like I feel like there's something I'm missing yeah that's uh I mean I think it's very interesting you point out that things that do not have cidal flow sort of endure over a longer period of time um I never thought about that okay okay fine fine fine fine I won't push it then okay that's really okay yeah it's interesting I need to think about it yeah yeah yeah it well God don't know what kind of worms I've just opened up I really don't want to get shouted out on the comments um okay so something I do understand a little bit better is this uh KO Divergence uh that's more up my streaks I feel like it's a little bit more cognitive sciencey so it's coolback Lio Divergence and and as I said this Um this can be cast as a distinction if we're doing basian inference between a true posterior and what's called the recognition density um which is like your version of your of of a posterior now there was something you bought up in that uh or car Bop or whoever wrote it in that um simplifying the free energy principle paper which is called a Fisher information metric tensor which measures

changes in the K Divergence it might be so people I think view the variational for energy as a kind of snapshot in time so given like this perceptual inference how big is my K Divergence given my action how much surprise is there but this seems to me to integrate change over time so what is the Fisher information metric tensor and how can we integrate that into a more accurate or more faithful model of the world or model of ourselves right uh so once you accept that you have a recognition density about the world and that this is minimizing free energy you can ask how much does your recognition density or your beliefs about the world have to change in order to minimize free energy and so if you if you want to quantify how much your beliefs move in a certain amount of time you need what's called uh an information metric information metric in the sense that it uh quantifies how much information changes over time the information have about and the information the fishal information metric is what allows you to do that nice okay cool are you are you familiar with this notion that's been coming out of maybe like the more philosophical or cognitive science side of active inference so people like Mark Miller and Julian Kine of prediction error Dynamics um I mean intuitively I can see what that would mean but yeah the idea is that we have like a higher order prior that we will track how good we're doing at minimizing free energy over time and this constitutes and and this in terms of psychology maps onto something like positive or negative affect because you know we've got a great we're doing well right we're doing well we're finding these slopes of um Vari variational for energy that we can descend down and there kind of an explanation for why we don't seek out dark rooms so there's one other solution to the dark room problem it sounds quite similar to what you're saying which is except what you're saying seems strictly I I don't know in terms of the K Divergence tied to perceptual inference but we can say that a system is becoming better perceptual inference because over time those KO divergences are going down is that reasonable to say well yes and no because if if the data stream sort of like stayed constant oh yes or if you stop the data stream then your kale Divergence would go down because you would kind of like get a free energy minimum yeah but always have new data coming in so you have two forces at play you have you uh who are like predicting the data and minimizing free energy but then you have data that's potentially increasing the free energy right but is the world but going back to our long equation the world is the world is modelable in some sense it's predictable in certain senses so I I may be wrong but it seems to me that the more data you get over time the better your model is going to

be right it's it's just going to become more powerful right to use a statistical term what what's the kind of so that seems to me that like your information metric should be pointing that you know your K divergencies are going down even if you're getting more data because that data's just it's got a similar distribution to the data that came before right okay so so I see I see where where the confusion lies um I agree with you that as we get more data our model gets better and our free energy um in general tends to be lower so that's one truth the other truth is that even though even even though I mean that's a fact your the data that's coming in keeps changing uh which means that the posterior density the true the true keeps changing so your recognition Dynamics always have to keep tracking this posterior distribution right now the fish looks at how much these um recognition Dynamics have to travel in a certain period of time to match the true posterior is there an equation that governs that because obviously I've become obsessed with equations um is there is there how much your beliefs change no not just how much your beliefs change but kind of how the data changes how the data changes that's the longine equation long equation but it's so that's the function of the X being the worldly States yeah got it got it okay okay yes okay I see where my confusion was I guess I was viewing the world as kind of static I mean you can even like take away the r the the random noise right if your f of x is changing your agent who's modeling that is going to have to keep updating keep changing their model nice nice okay cool that's that's that's very useful okay excellent um okay have I got any more maths um I got a bit more maths I guess one one something I wanted to ask about is this is a broader question so in so in active inference we have discrete State spaces and continuous State spaces so and it's kind of about the how we carve up time in some sense so continuous State spaces is where it doesn't really make sense to put time into separate chunks or sequential chunks so you know if you're thinking about attention or perception or perception let's say um I know perception includes some action but let's just take perception it's better to kind of model that in terms of a continuous State space if you're thinking about decision- making you would think about that in terms of discrete State spaces because an action policy is a discrete thing in discrete state space we've seen this kind of proliferation of So-Cal pom DP schema um so these are these partially observable Markov decision blank but process close always think about Marco blankets um is there an equival of that in continuous State spaces uh so you can definitely have continuous uh parti obser Mark of decision processes um so so let let me take a step back there's two things that play there's a

whether the stage space is discrete or continuous and whether time is discrete or continuous okay now in active inference implementation so far you basically have either continuous space continuous time or discrete space discrete time yeah yeah or both but people haven't really been playing with continuous space discrete time right uh continuous time discrete space but there's no reason why in principle you would not be able to do that okay that's cool well uh this is again my very much lay and knowledge of physics um do we not take time and space to be somewhat interwoven um yeah this is a bit beyond me I guess I guess I guess a way around this is to say that we're modeling right so in our modeling we can take time as continuous and space as discrete and vice versa yeah this is this is only about modeling about how you represent the world but there is a deeper uh a deeper question which is how do we ultimately like represent the world in physics and it's true that some theories you know consider time and space jointly but for us humans uh yeah we don't need to get into that for active res modeling okay so we we've we've kind of clarified that so when so where would the pomdp schemes be able to be used if it's not just discrete time discrete space they applicable to continuous space continuous time so so where would you actually use them so I've seen for example just to give some background I've kind of seen pom DP schemes used in mainly just aspect like context of decision making so um for example yeah going back to the flow paper flow involves selecting action policies meditation involves mental action um attention right like um intentional uh attention uh endogenous attention is a decision these can all be modeled quite nicely but to my eyes everyone who talks about perception will use something like a predictive coding scheme yeah um is there is that I'm just curious about what is if there is an alternative to that in the literature in the literature no okay I was talking to Ryan Smith recently and he was interested in continuous State P DP with discrete and so the idea here is that you can Model A continuous representation of the world and a sequential decision making at the same time of that so possibilities okay cool I'm speaking to Ryan fairly soonish so that that would be useful that would be cool that's a good thing to bring up yeah that's really interesting I don't know this brings to mind there's this incredible paper by a guy called Juan Diego botar and I think his name is Zacharia Jabara and it's sort of symphysis subjective time and objective time through a basian model and I'm going to tell you I stared to that graph for about 6 hours um and it still did make sense so that's a you know if anyone wants to read that and try and think about how maybe not space and time but like two different types of time can be coals

that's a good place to start so I'll give you another theory of mine uh I attended this wonderful work workshop on computational neurop phenomenology uh last year and I had a wonderful conversation with annel Seth and we kind of converged onto the hypothesis that subjective time is uh the amount of distance that your beliefs travel over some period of objective time okay cool so here here's the thing like when you when your beliefs move a lot yeah and you can quantify that with the Fisher information metric you would have a perception that a lot of time has elapsed okay so you've now got me on this new pet hobby of mine because I'm sure you haven't seen it but literally yesterday I published a preprint called distrusting the policy how inference over action shapes our perception of time that's cool I should read so so please I would love some feedback basically I wrote it so and this this started off as being trying to explain temporality in flow States and I thought no actually this works I think this works more broadly so I'm not going to spoil it well I will spoil it but basically my idea so I'm building off what yakob Harry called distrusting the present so this is very similar to what you're saying so yakob Harry has this idea that this the flow of time is contingent on um the rate at which we update our perceptual hypothesis about the world so if I'm in a volatile environment I the uh the state estimation I have at time T is not going to work for t plus one and so what happens is the moment I find T I immediately start discounting it which you can consider sort of precision waiting and immediately I have to update that and he C that now I I don't think that that totally maps onto the phenomenology because I'm really curious about what you think about this and again I will try and do it quickly because I don't want to bore you but we have um so I say so he was kind of it seems to me he was rooting uh flow of time in predictability but the problem is is if you're going to cast it only in terms of perception or only in terms of the a matrix then you you've got a slight problem because there are two different types of predictability you've got one which I call static predictability which is basically when you're staring at a wall now when you're staring at a wall your perceptual hypothesis is going to be pretty good let's say over a prolong period of time and so Jacob's Theory says time will slow down and it does right you stare at a wall for an hour it's going to feel like a day yeah but then the other problem is something called Dynam what I call Dynamic predictability so let's say you've played a video game level once it's very exciting time seems to flow quickly now let's say I make you play that same video game a 100 times now perceptually the world is changing you got to do all of that state like that state estimation that combination of the

D Matrix and the A matrix needs to keep happening at the same rate and I want to flesh this out experimentally but I have a hunch that time is going to move way slower so what I've what I've now said is it's about the frequency with which you update your action policies and the reason why I said that is because with each action policy updating you also have to update your beliefs in the C Matrix and the C Matrix is your expected pre uh your preferred sensory observations and that I think accounts for the perceptual aspect of the flow of time that's really that's a really interesting hypothesis yeah yeah so I so it's Friday the 19th of January on Tuesday people are not going to be able to you know this is not people can't go back in time anyway I'm doing a talk about it at the at the institute on Tuesday um so anyway that that's so I think we're converging on similar points here I think it's kind of where is the source of your information change so for me it's the C Matrix really because it can't be the B Matrix because in Dynamic predictable so uh B Matrix of State transitions in dynamically predictable situations like playing the same video game 100 times I'm going to have the same number of B matrices compared to when I played it the first time so I don't think it's the B Matrix but because I you have to keep doing this expected free energy calculation for each action policy and the C Matrix feeds into that my hunch is that it might be rooted than that right you you could also think about the fact that if you replay the video game uh like many times mhm your belief still have to move to adapt to the current uh situation in the video game but there will be less and less novelty yeah because you adapted your like big world model to like predict the video game very so how that translates to in practice is that you have lowlevel beliefs that that move a lot to you adapt to current situation but high level beliefs that don't move because okay nice everything again I think I'm trying to get this to be done experimentally to see if it's if this Theory maps onto anything yeah it would be really nice to test but that's that's a very interesting yeah I'm always in favor of all all of the empirical experiments I feel as a community we we're lacking that I think that's right and I think it shouldn't be that hard to do I think you just have to give people statically predictable context dynamically predictable context and flow context yeah I mean um as soon as you can map a p onto the data that's coming in be very easy to model everything that's going on well I may reach out to you or LS on that one um Okay cool so that that was a fun little for um okay actually let's let's um we we spoke we very briefly touched upon attention um and I have a question about attention if I may i' I've been I've been I've been grappling with attention for quite a long time um so attention is kind of

construed as optimization of the likelihood distribution in active inference of the a matrix in the pomdp schema now there are these really lovely papers by his name is mitah so uh mital 2016 2018 and 2019 I think and he shows he gives this example of like searching for a red pen and he says that when you're searching for a red pen what you do is you downweight sensory Precision of stimuli that are Rel irrelevant to that context and so basically you end up doing Salient Behavior which is we can view as epistemically Salient because it yields Information Gain but the information that you want um now given that um yeah so that makes sense because task relevant stimuli are downweighted so only task relevant stimuli involved in this maximal Information Gain okay fine but pragmatically so so that epis that information gain is like I didn't know where the red pen was now I know where the red pen is yeah but in terms of pragmatic in terms of pragmatics we know that action policies in uh Act of influence are driven by a kind of almost a false belief a false preference so I might say I have a red pen and I'm going to go out there to confirm that belief because without it there's lots of prediction error or variation or free energy this seems to be a kind of issue because on one hand I'm gaining information about the world so I kind of knew that I didn't have a pen and now I know where the pen is and so my I have this K Divergence but then I also have this oddly pragmatic belief that I do have a red pen because without that red without that belief I wouldn't be motivated to go and get the red pen so I'm I'm trying to Grapple with how to resolve that clash and I'm curious if you can help yeah not sure um but what I'm thinking is uh I always think about attention as control of the a Matrix so kind of like in large large's paper on meditation mhm uh you make some some uh sensory modalities more precise and others less precise yeah and it feeds into into your computations yes so yeah see I've always I've always found it quite useful to distinguish Precision from Precision waiting so Precision I just view as the kind of inverse entropy or of a distribution but Precision waiting is actually what I do right I wait precision and that's not all going to be down weirdly Precision waiting is not all down to Precision well isn't it I think it often is it often is in terms of so they talk about how attention is basically one definition of it is the capacity to kind of sift through the data that you currently have for the most uh informationally reliable or precise um data right so the one that's going to say okay given this I know exactly what caused it but clearly like for example in my searching of the red pen there's going to be loads of things in the environment that are that have a very uh you know very strict uh relationship mapping between the

observation and the state so while I'm scanning I might see something and go oh obviously that's my phone right it's not ambiguous whether it's my phone or not but because I'm but the whole point of this Mit paper is I don't I don't care so I'm downweighting that stimuli but that's nothing to do with how well it maps onto some State yeah yeah it isn't okay so if I understand correctly this is um because you you want to conform to your prior belief and therefore if your prior belief is very precise mhm uh you will discount whatever sensory information comes in right so my prior belief is yeah so my prior belief is something like I have a red pen and and and so that's very precise and yeah so that leads you to Discount things that are because you're you're you're yeah you're only going to pay attention to things that are task relevant yeah then so but that but that's kind of but the point is that's kind of a belief saying I have a pen but then there still is this epistemic gain which is at the heart of Sal which is I didn't have a pen or I didn't know where the pen was and now I do know where the pen is and so you're starting with I don't know where the pen is but for your act to Trigger action you have the belief I know where the pen is to to Trigger action to look for the pen I'm I'm getting I'm getting a little bit lost okay so so um preferences underly action right so a see Matrix so I have the preference about the sensory information that I expect to receive mhm so let's say and and preferences can be casted as just beliefs so I believe so for example uh in some very it's it's some very subpersonal way I believe that uh I'm going to be at 37.9 degrees and so I act so I remain around that now I want to find my red pen we can cast that valtion as a preference as a belief yeah I know where my red pen is and the reason why you would do that is because well the world is not giving you that information and so you've got prediction error and you want to minimize that prediction error so that triggers action to confirm the belief that you have a red pen okay that's on one side but on the other side what captures attention is that which is most Salient and what salience can be cast as is epistemic gain Information Gain and that for me in this red pen example looks like well I didn't know where the red pen was and now I do know where the red pen is but what you've got on one hand is on the epistemic side you've start with the belief I don't know where the pen is because if I did know where the pen is finding the pen would not be very useful right so that's um inhibition of return that kind of explains inhibition of return that's on the epistemic side on the pragmatic side I have I have a pen so I'm just asking how those two things come together I'm I'm so sorry Darius I think my brain is fried by that's all right that's right I'm sure someone in the comments will

have an answer yeah I kind of feel that KL will just have an answer um I mean I'm sure I'm sure he has an answer to everything Maybe not maybe not okay let's um a research paper yeah maybe I I know that one's going to take me down some wormholes um so okay last thing I guess I want to talk about is you went on the machine learning Street Talk podcast and you spoke a lot about core knowledge which I thought was really interesting I really didn't expect that to come up um but I know it's one of your interest terms of what are the kind of priors that need that are built that facilitate intelligence sentient Behavior so I guess what are your current thoughts on how so it's interesting because I actually come from a Linguistics background as I told you before we started and at least you know at Oxford where I was they they love Chomsky so Chomsky has this notion of universal grammar there's something built into the genetic code that facilitates language learning so I'm very familiar with these arguments Poverty of stimulus and these arguments in terms of nativism what are your thoughts on how fleshed out those priors are um in terms of being innate well I think there's definitely innate priors when we were born and this is all studied um I think one of the best people is Elizabeth Spelke y her whole career uh describing and characterizing those priors MH uh these seem to be incredibly precise uh much like we cannot change them and they seem to be super General in the sense that they would apply to any kind of naturalistic or physical environment that you could ever be in so these are things like object permanence two object cannot uh interact at a distance but two people can interact at a distance so so two agents can uh I think there's a core knowledge system for um numbers learning numbers there might also be a core knowledge system for language uh they might also be one for vision uh so there's six of them and she wrote a book about it uh which is what babies know uh that I absolutely recommend with with all of that research so it's a big research program to actually being able to understand those and reveal those core systems it's another one to be able to reverse engineer them interesting yeah I think one of the issues I guess is at least um in the history of linguistics one of the great problems was how rich are these priors so this is what later got known as Darwin's problem uh because Chomsky has this idea that well in the 80s he had this principles and parameters idea which is that what we have is a bunch of parameters that can be switched on or off uh you know given sensory data those parameters get switched on or off and then the problem is you had this massive proliferation of parameters so there were 500 parameters and then finished people you know there was one only for finish

but obviously we all had it but only unless you were exposed to finish did you ever turn it on and it just became completely inexplicable given Evolution because how on Earth did one mutation or a couple of mutations trigger that so he now thinks that there's something a lot more paired down something like recursion and I think why why that's interesting is because recursion is not is not what we call propositional Knowledge it's not knowledge that something so Trump might say oh okay you know in the 60s might said okay I know the baby knows that after this comes this or something but now this is just actually a mechanism so what we might call procedural knowledge when you think about uh core knowledge for maths or call knowledge for vision or I guess Vision really won't be propositional call knowledge for language are you construing these as propositional or procedural um so well these would be priories in the G of model which could be expressed um in a number of ways it could be uh sort of like um like a grammar of uh like core components that you already have in your G model well well let let me put it in a different way like when we're born we have a very simple gen model and gradually we learn about the RO around this and we make it more complex now what we have at the beginning is is this core knowledge it's kind of like a starting point Gant model that allows you to right by structure learning learn about everything else now the structure of that is uh very debated among computational people so for example Josh tenenbom is pursuing this idea of probabilistic programs which are B based on symbolic grammars and probabilistic rules between different entities then there's other people who might uh model these things with neural networks right so in those cases you don't have symbols you don't have that grammar going on yeah it's very much an open area of research cool yeah I mean I guess all these neural networks still have some priors um in in a lot of cases yeah awesome yeah yeah I used to love these debates between Chomsky and people like um Jeffrey Elman sort of mid 1990s proper hardcore stuff cool and yeah I I it's interesting you mention Vision because I always thought that for me the one glaring one where there needs to be PRI is Vision vision is perception is well visual perception is such an amazing capacity yeah to be able to there is a startup called a Common Sense machines MH has pushed the core knowledge of vision really to state-of-the-art level where they have a world model for vision and then they do inference on that world model and they they're able to reconstruct 3D scenes from 2D images wow in in a pretty amazing way so yeah out have you seen these things uh the this is slightly different but you have these neuron Nets that are trained on like uh brain Imaging data and they can

take an MRI of the brain and say that this person was thinking about a dolphin or something they can have an image of what that person is thinking about obviously we're given the training that you know there is this I'm not going to say causation correlation I won't go into it but that you know when this part when this brain data is like this this person is thinking this and so you have the testimonies of the people but it just shows the power of those and the and the kind of nuance of those large language models that they can be able to do that um but but actually I think it's arguably less exciting than people think it is because they forget about the training process that underlies it yeah and it doesn't allow you to understand it just allows you to predict exactly exactly and that's where that's where our our favorite topic of active inference might might come in maybe and yield some artificial understanding hopefully hope well people are getting very excited getting very excited L that was that was awesome I feel like I've learned I feel like I've gone to University again yeah me too no this no this was so much fun I knew it would be fun thank you for being gentle with me and kind with the maths well thanks for having me no it was an absolute pleasure um I really want to see what comes out of this stuff especially with annel about the Fisher information metric and time yeah is g to come out really cool yeah cool sounds like we're converging potentially on a similar idea so yeah it would be really nice to talk I need to flesh out these ideas and then yeah we can talk more Splendid okay well thank you from longan and [] PL and all of this stuff it's been an absolute pleasure so thank you for myself and The Institute it's been a great pleasure for me too thanks